

Cancellation of Self-sensing due to Ego-motion of Vibration based Mechanosensors

1. Introduction

Tactile sensing is an important capability in both biological organisms and autonomous robots. Many insects use antennae as active tactile sensors to detect obstacles, recognize surfaces, and navigate through their environments. Inspired by these biological systems, researchers have developed vibration-based robotic antennae using flexible probes and accelerometers. These systems are especially useful in low-light or visually difficult environments where cameras are less effective. However, when the robot moves, vibrations generated by the robot itself create false sensor readings. This effect is called self-sensing or ego-motion disturbance. The main objective of current research is to separate true contact signals from self-generated vibrations using signal processing and machine learning techniques.

2. Biological Inspiration of Robotic Antennae

According to Staudacher et al. (2005), insect antennae are highly specialized tactile sensing organs equipped with mechanoreceptors and proprioceptors. These structures allow insects to actively explore nearby objects, identify obstacles, and understand environmental properties. The review explains that insects such as stick insects, crickets, cockroaches, locusts, and honeybees use antennae for active sensing during locomotion. The antennae can move in different directions using joints and muscles, allowing wide spatial coverage.

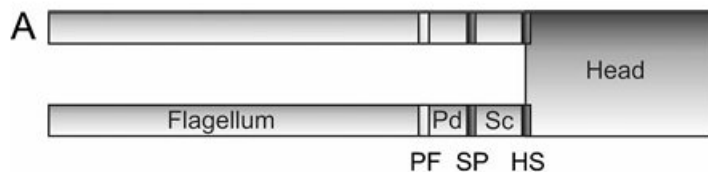


Figure 1: Schematic top view of an insect head. The antennae consist of three functional segments: the scape (Sc), pedicel (Pd) and an annulated flagellum.

The antenna typically consists of three main parts: the scape, pedicel, and flagellum. The flagellum acts as the primary sensing region. When the antenna touches an object, mechanical vibrations travel through the structure and can be measured using mechanoreceptors. The same principle is used in artificial robotic antenna systems where an accelerometer is mounted at the tip of a flexible probe.

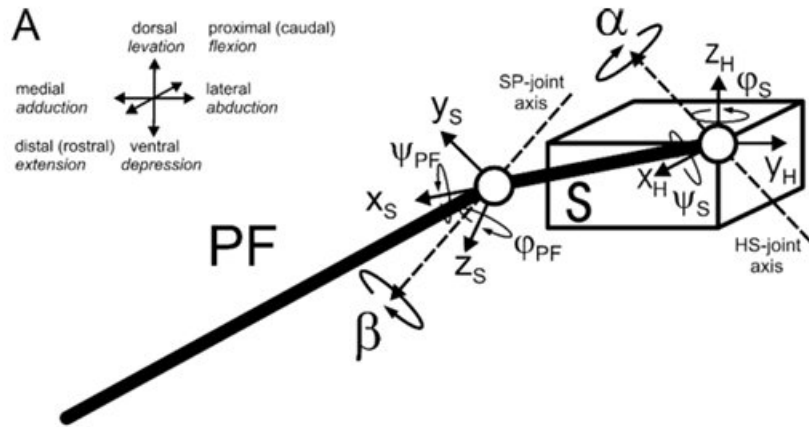


Figure 2: Kinematic Model of Insect Antenna Movement

The paper also explains that insects can distinguish between self-generated motion and external touch by combining motor information with sensory feedback. This concept directly inspired the development of forward models and bio-inspired algorithms used in modern robotic tactile systems.

- Block diagram of sensor → signal processing → machine learning → touch detection.

3. Review of Main Research Papers

Paper [1]– A Forward Model for an Active Tactile sensor using Echo State Networks

This study introduced a forward model based on Echo State Networks (ESN) to predict self-generated vibration signals in a robotic antenna system. The robotic antenna consisted of a flexible plastic tube with an accelerometer mounted at the tip. The antenna was attached to a mobile robot using a pan-tilt mechanism. Inputs to the ESN included robot acceleration data and servo motor position signals. The study showed that ESNs can successfully predict sensor readings during robot movement, thereby enabling the identification of external touch events.

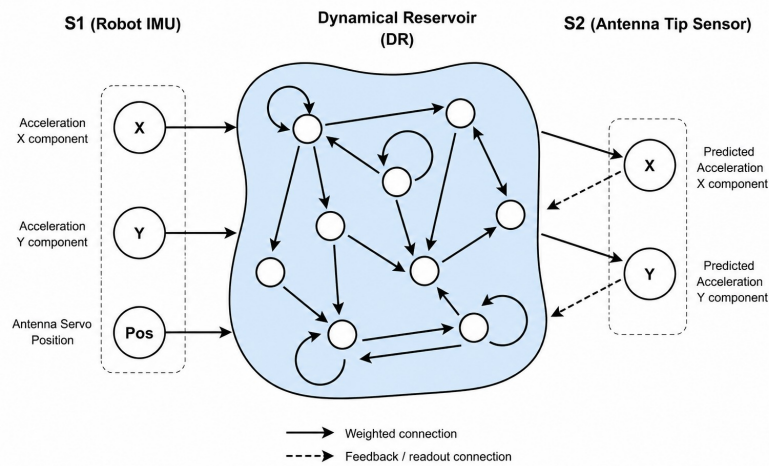


Figure 3: Echo State Network showing the inputs, S1 and antennal servo position (Pos), Dynamical reservoir (DR), and the output S2 (signal to be predicted). X and Y represent the two components of the acceleration vector in each sensor.

Paper [2] – Prediction of Self-induced Mechanoreceptive Sensor Readings in an Insect-inspired Active Tactile Sensing System

This paper expanded the previous work by introducing two antennae with multiple motion patterns. Experiments included straight robot motion, curved motion, and rhythmic disturbances created using spiked wheels. The researchers used signal rectification and low-pass filtering to improve touch event detection. The study demonstrated that forward models can significantly reduce false detections caused by ego-motion.

Paper [3] – Distance Estimation and Material Classification of a Compliant Tactile Sensor Using Vibration Modes and Support Vector Machine

This research focused on using vibration analysis for estimating contact distance and identifying material types. FFT analysis, Artificial Neural Networks (ANN), and Support Vector Machines (SVM) were compared. The researchers observed that vibration

characteristics change depending on the contact location along the antenna. SVM-based approaches provided improved performance for classification tasks.

4. Comparison of Reviewed Studies

Research Paper	Main Technique	Sensors Used	Main Objective	Advantages	Limitations
[1]	Echo State Network	Accelerometer + Servo Position	Predict self-generated vibrations	Good prediction capability	Requires parameter tuning
[2]	ESN + Filtering	IMU + Accelerometers	Detect true touch events	Works under multiple motions	Complex preprocessing required
[3]	FFT + ANN + SVM	MPU6050 Accelerometer	Distance & material estimation	High classification accuracy	Reduced accuracy near antenna tip

Although existing studies successfully reduced self-generated sensor disturbances, several challenges still remain. Real-time processing is computationally demanding, especially when multiple sensors are used. The performance of machine learning models depends heavily on training conditions and parameter selection. Furthermore, many systems are tested only in controlled laboratory environments. More research is needed to improve robustness under real-world conditions such as uneven terrain, continuous robot motion, and environmental noise.

Another important challenge is adaptive learning. Most current systems use pre-trained models and fixed thresholds. Future systems should automatically adapt to changing conditions and robot dynamics. Lightweight embedded implementations are also necessary for practical mobile robotic applications.

5. Conclusion

The reviewed literature shows that bio-inspired tactile sensing systems are highly effective for robotic obstacle detection and environmental interaction. Researchers have successfully used vibration analysis, signal processing, and machine learning to reduce self-sensing caused by robot motion. Echo State Networks, FFT analysis, ANN, and SVM methods have shown promising results for predicting and classifying tactile signals. These studies provide a strong foundation for the current project on cancellation of self-sensing due to ego-motion of vibration-based mechanosensors.

8. References

- [1] Harischandra, N., & Dürr, V. (2012). A Forward Model for an Active Tactile Sensor Using Echo State Networks.
- [2] Harischandra, N., & Dürr, V. (2013). Prediction of Self-induced Mechanoreceptive Sensor Readings in an Insect-inspired Active Tactile Sensing System.
- [3] Gunasekara, S. R., et al. (2019). Distance Estimation and Material Classification of a Compliant Tactile Sensor Using Vibration Modes and Support Vector Machine.
- [4] Staudacher, E. M., Gebhardt, M., & Dürr, V. (2005). Antennal Movements and Mechanoreception: Neurobiology of Active Tactile Sensors.